Complex Numbers

OVERVIEW

1. Exponential form

Write $z = re^{i\theta}$.

2. Products and quotients

Multiply moduli and add arguments; divide moduli and subtract arguments.

3. De Moivre's theorem

Find powers of complex numbers.

4. Trigonometric identities

Connect multiple angles and powers of sine and cosine.

5. Complex series

Use geometric series and compare real and imaginary parts.

6. Roots

Find roots and interpret regular polygons on Argand diagrams.

RECAP: Modulus-Argument Form

Solution 3

For $z = x + iy$, let $r = |z|$ and $\theta = \arg z$.

Find expressions for r, x, y.

Coordinates

$$r = \sqrt{x^2 + y^2}, \quad x = r\cos\theta, \quad y = r\sin\theta.$$

Modulus–argument form

$z = r(\cos\theta + i\sin\theta)$.

The principal argument satisfies $-\pi < \theta \leq \pi$.

Use the quadrant as well as $\tan\theta = y/x$. In the first quadrant, $\theta = \tan^{-1}(y/x)$.

The argument of zero is undefined.

RECAP: Modulus-Argument Form

Solution 4

Write -1 in modulus–argument form.

a

$r = 1,\ \theta = \pi.$

$z = \cos\pi + i\sin\pi.$

Write i in modulus–argument form.

b

$r = 1,\ \theta = \pi/2.$

$z = \cos\dfrac{\pi}{2} + i\sin\dfrac{\pi}{2}.$

(c) $1 + i$

c

$r = \sqrt{2},\ \theta = \pi/4.$

$z = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}).$

(d) $-\sqrt{3} + i$

d

$r = 2,\ \theta = 5\pi/6.$

$z = 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}).$

Exponential Form — Solution 5

Euler's formula

Recall the Taylor expansions:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Substitute $x = i\theta$ into the exponential series.

Substitute

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \cdots$$

Group

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \cdots\right).$$

Identify

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

Exponential Form — Solution 6

Convert to exponential form

(a) -1

a $-1 = e^{i\pi}$. Hence $e^{i\pi} + 1 = 0$.

(b) $2 - 3i$

b $\sqrt{13}\, e^{-i\tan^{-1}(3/2)}$.

(c) $\sqrt{2}(\cos\frac{\pi}{10} + i\sin\frac{\pi}{10})$

c $\sqrt{2}e^{i\pi/10}$.

(d) $-1 + i$

d $\sqrt{2}e^{3i\pi/4}$.

(e) $2(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5})$

e $2e^{3i\pi/5} = 2e^{23i\pi/5}$.

Arguments differing by $2k\pi$ describe the same number. Choose the principal argument when requested.

A Final Example

Solution 7

Show that

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}.$$

Answer

$e^{i\theta} = \cos\theta + i\sin\theta,$

$e^{-i\theta} = \cos\theta - i\sin\theta.$

Add the equations and divide by 2:

$$\boxed{\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}}.$$

Exercise 1A - Core Pure Year 2 Page 5

Solution 8

Exercise 1A · Question 1

Express in the form $re^{i\theta}$, with $-\pi < \theta \leq \pi$. Use exact values where possible, otherwise 3 significant figures.

(a) -3

a $r = 3,\ \theta = \pi$, so $3e^{i\pi}$.

(b) $6i$

b $r = 6,\ \theta = \pi/2$, so $6e^{i\pi/2}$.

(c) $-2\sqrt{3} - 2i$

c $r = \sqrt{12 + 4} = 4.$

The point is in the third quadrant:

$$\theta = -\pi + \tan^{-1}\frac{1}{\sqrt{3}} = -\frac{5\pi}{6}.$$

$z = 4e^{-5i\pi/6}.$

Exercise 1A · Question 2

Express in the form $x + iy$, where $x, y \in \mathbb{R}$.

(a) $e^{i\pi/3}$

a $\frac{1}{2} + \frac{\sqrt{3}}{2}i.$

(b) $4e^{i\pi}$

b $-4.$

(c) $3\sqrt{2}e^{i\pi/4}$

c $3 + 3i.$

(d) $8e^{i\pi/6}$

d $4\sqrt{3} + 4i.$

(e) $3e^{-i\pi/2}$

e $-3i.$

(f) $e^{5i\pi/6}$

f $-\frac{\sqrt{3}}{2} + \frac{1}{2}i.$

Exercise 1A · Question 3

Express as $r(\cos\theta + i\sin\theta)$, with $-\pi < \theta \leq \pi$.

(a) $e^{16i\pi/13}$

a $16\pi/13 - 2\pi = -10\pi/13.$

$\cos(-10\pi/13) + i\sin(-10\pi/13).$

(b) $4e^{17i\pi/5}$

b $17\pi/5 - 4\pi = -3\pi/5.$

$4[\cos(-3\pi/5) + i\sin(-3\pi/5)].$

(c) $5e^{-9i\pi/8}$

c $-9\pi/8 + 2\pi = 7\pi/8.$

$5[\cos(7\pi/8) + i\sin(7\pi/8)].$

Multiplying and Dividing Complex Numbers

Solution 11

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$.

Multiply

Product

$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$.

Multiply the moduli; add the arguments.

Trigonometric product

$z_1 z_2 = r_1 r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]$.

Divide $\cdot\ z_2 \neq 0$

Quotient

$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$.

Divide the moduli; subtract the arguments.

Trigonometric quotient

$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)]$.

Examples

Solution 12

1. Multiply

$3(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12})$
$\times 4(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12})$.

1

$12e^{i\pi/2} = 12i$.

2. Multiply

$2(\cos\frac{\pi}{15} + i\sin\frac{\pi}{15})$
$\times 3(\cos\frac{2\pi}{5} - i\sin\frac{2\pi}{5})$.

2

$6e^{-i\pi/3} = 3 - 3\sqrt{3}i$.

3. Write in exponential form

$$\frac{2(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12})}{\sqrt{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})}.$$

3

$$\sqrt{2}e^{i(\pi/12-5\pi/6)} = \sqrt{2}e^{-3i\pi/4}.$$

Tip

$\cos\theta - i\sin\theta = \cos(-\theta) + i\sin(-\theta)$.

Test Your Understanding — Solution 13

Edexcel FP2 June 2013 · Q2

$z = 5\sqrt{3} - 5i$. Find (a) $|z|$ (1 mark); (b) $\arg z$, in terms of π (2 marks).

a $|z| = \sqrt{75 + 25} = 10.$

b

$\arg z = -\tan^{-1}(1/\sqrt{3}) = -\pi/6.$

$w = 2(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$.

Find (c) $|w/z|$ (1 mark); (d) $\arg(w/z)$ (2 marks).

c $|w/z| = 2/10 = 1/5.$

d

$\arg(w/z) = \pi/4 + \pi/6 = 5\pi/12.$

Exercise 1B - Core Pure Year 2 Pages 7-8 — Solution 14

Exercise 1B · Question 6

Use exponential form to show that

$$\frac{(\cos 9\theta + i\sin 9\theta)(\cos 4\theta + i\sin 4\theta)}{\cos 7\theta + i\sin 7\theta}$$

$\equiv \cos 6\theta + i\sin 6\theta.$

Answer

$$\frac{e^{9i\theta}e^{4i\theta}}{e^{7i\theta}} = e^{(9+4-7)i\theta}$$

$= e^{6i\theta} = \cos 6\theta + i\sin 6\theta.$

Exercise 1B - Core Pure Year 2 Pages 7-8 — Solution 15

Exercise 1B · Question 7

$z = 1 + i\sqrt{3}$, $\mathrm{Re}(z^2/w) = 0$ and $|z^2/w| = |z|$.

Use geometrical reasoning to find both possibilities for w in exponential form. (4 marks)

Answer

$z = 2e^{i\pi/3}$. The modulus condition gives $|w| = 2$.

Since z^2/w is purely imaginary, its argument is $\pm\pi/2$.

$$\frac{2\pi}{3} - \arg w = \pm\frac{\pi}{2} \pmod{2\pi}.$$

Hence

$$\boxed{w = 2e^{i\pi/6} \text{ or } 2e^{-5i\pi/6}}.$$

Exercise 1B - Core Pure Year 2 Pages 7-8 — Solution 16

Exercise 1B · Question 8

(a) Evaluate $(1+i)^2$ in exponential form.

a $1 + i = \sqrt{2}e^{i\pi/4}$, so $(1+i)^2 = 2e^{i\pi/2}$.

(b) Prove by induction that

$(1+i)^n = 2^{n/2}e^{in\pi/4}, \quad n \in \mathbb{Z}^+$.

b

True for $n = 1$. Assume true for $n = k$. Then

$$(1+i)^{k+1} = (1+i)2^{k/2}e^{ik\pi/4}$$
$$= 2^{(k+1)/2}e^{i(k+1)\pi/4}.$$

Thus it holds for $k+1$, so for every positive integer by induction.

(c) Hence find $(1+i)^{16}$.

c

$2^8 e^{4i\pi} = 256$.

Exercise 1B - Core Pure Year 2 Pages 7-8 — Solution 17

Exercise 1B · Question 9

Use Euler's relations for $e^{i\theta}$ and $e^{-i\theta}$ to verify $\cos^2\theta + \sin^2\theta \equiv 1$.

Answer

$e^{i\theta} = \cos\theta + i\sin\theta,$

$e^{-i\theta} = \cos\theta - i\sin\theta.$

Multiply:

$1 = (\cos\theta + i\sin\theta)(\cos\theta - i\sin\theta)$

$= \cos^2\theta + \sin^2\theta.$

Challenge 1B - Core Pure Year 2 Pages 7-8 — Solution 18

Challenge 1B

(a) Prove by induction, for positive integers n, $(re^{i\theta})^n = r^n e^{in\theta}$.

Use only the complex multiplication rule, not an assumed law of indices.

a

True for $n = 1$. Assume true for k.

$(re^{i\theta})^{k+1} = re^{i\theta}\, r^k e^{ik\theta}$

$= r^{k+1}e^{i(k+1)\theta}.$

Hence true for all positive integers by induction.

(b) Given $z^{-n} = 1/z^n$, show that $(re^{i\theta})^{-n} = r^{-n}e^{-in\theta}$ for $r > 0$. Use the complex division rule.

b

$$(re^{i\theta})^{-n} = \frac{1}{r^n e^{in\theta}}$$

$= r^{-n}e^{i(0-n\theta)} = r^{-n}e^{-in\theta}.$

De Moivre's Theorem

Solution 19

De Moivre's theorem

If $z = r(\cos\theta + i\sin\theta)$, what is z^n?

Theorem $z^n = r^n(\cos n\theta + i\sin n\theta)$.

Prove the result by induction for positive integers n.

Edexcel FP2 June 2013 · Q4

Induction

True for $n = 1$. Assume true for $n = k$.

Multiplying by z, the product rule gives

$z^{k+1} = r^{k+1}[\cos(k\theta + \theta)$

$+ i\sin(k\theta + \theta)]$.

This is the result for $k + 1$; therefore true for all positive integers.

Alternative

Alternatively, expand the product using the angle-addition identities:

$\cos k\theta\cos\theta - \sin k\theta\sin\theta = \cos(k+1)\theta$,

$\sin k\theta\cos\theta + \cos k\theta\sin\theta = \sin(k+1)\theta$.

De Moivre's Theorem for Exponential Form

Solution 20

If $z = re^{i\theta}$, find z^n.

Answer

$z^n = (re^{i\theta})^n = r^n e^{in\theta}$.

Raise the modulus to the power n; multiply the argument by n.

For negative integers, $z \neq 0$.

Examples

Solution 21

1. Simplify

$\dfrac{(\cos\frac{9\pi}{17}+i\sin\frac{9\pi}{17})^5}{(\cos\frac{2\pi}{17}-i\sin\frac{2\pi}{17})^3}$.

1

$$\frac{e^{45i\pi/17}}{e^{-6i\pi/17}}=e^{51i\pi/17}=e^{3i\pi}=-1.$$

2. Express in the form $x+iy$

$(1+\sqrt{3}i)^7$.

2

$1+\sqrt{3}i=2e^{i\pi/3}$.

$(1+\sqrt{3}i)^7=128e^{7i\pi/3}$

$=128(\frac{1}{2}+i\frac{\sqrt{3}}{2})=64+64\sqrt{3}i$.

Test Your Understanding

Solution 22

Edexcel FP2 June 2010 · Q4

$z=-8+8\sqrt{3}i$.

(a) Find the modulus and argument of z. (3 marks)

(b) Use De Moivre's theorem to find z^3. (2 marks)

a

$|z|=\sqrt{64+192}=16$.

The point lies in quadrant II, so $\arg z=2\pi/3$.

b

$z^3=16^3e^{2\pi i}=4096$.

Exercise 1C · Question 5

Express $(3+i\sqrt{3})^5$ as $a+bi\sqrt{3}$, where a, b are integers.

Answer

$|3+i\sqrt{3}| = 2\sqrt{3}, \quad \arg(3+i\sqrt{3}) = \pi/6.$

$(3+i\sqrt{3})^5 = (2\sqrt{3})^5 e^{5i\pi/6}$

$= 288\sqrt{3}(-\frac{\sqrt{3}}{2} + \frac{i}{2})$

$= \boxed{-432 + 144i\sqrt{3}}.$

Exercise 1C · Question 6

$w = 2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$.

Find w^4 exactly in the form $a+ib$.

6

$w^4 = 16e^{2i\pi/3} = 16(-\frac{1}{2} + i\frac{\sqrt{3}}{2})$

$= \boxed{-8 + 8i\sqrt{3}}.$

Question 7

$z = \sqrt{3}(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4})$.

Find z^6 exactly in the form $a+ib$.

7

$z^6 = 27e^{-18i\pi/4} = 27e^{-i\pi/2} = \boxed{-27i}.$

Exercise 1C · Question 8

(a) Express

$$\frac{1+i\sqrt{3}}{1-i\sqrt{3}}$$

as $re^{i\theta}$, with $r > 0$ and $-\pi < \theta \leq \pi$.

a

$$\frac{2e^{i\pi/3}}{2e^{-i\pi/3}} = e^{2i\pi/3}.$$

(b) Find the smallest positive integer n for which this number raised to the power n is real and positive.

b

$$e^{2ni\pi/3} > 0 \iff \frac{2n\pi}{3} = 2k\pi.$$

Thus $n = 3k$; the smallest positive integer is $\boxed{3}$.

Exercise 1C · Question 9

Use De Moivre's theorem to show that

$(a+bi)^n + (a-bi)^n$

is real for integer n, wherever the powers are defined.

Answer

Write $a + bi = re^{i\theta}$, so $a - bi = re^{-i\theta}$.

$$\begin{aligned}(a+bi)^n + (a-bi)^n &= r^n(e^{in\theta} + e^{-in\theta}) \\ &= 2r^n \cos n\theta,\end{aligned}$$

which is real. For negative n, require $r \neq 0$.

Challenge 1C - Core Pure Year 2 Pages 10-11

Solution 27

Challenge 1C

Without Euler's relation, prove for positive integers n and $r > 0$:

$[r(\cos\theta + i\sin\theta)]^{-n}$

$= r^{-n}[\cos(-n\theta) + i\sin(-n\theta)]$.

You may assume De Moivre's theorem for positive integers.

Answer

By De Moivre and the definition of a negative power,

$$\frac{1}{r^n(\cos n\theta + i\sin n\theta)}.$$

Multiply numerator and denominator by the conjugate:

$$\frac{\cos n\theta - i\sin n\theta}{r^n(\cos^2 n\theta + \sin^2 n\theta)}$$

$= r^{-n}[\cos(-n\theta) + i\sin(-n\theta)]$.

Applications of de Moivre #1: Trig identities

Solution 28

Express $\cos 3\theta$ in powers of $\cos\theta$.

How can De Moivre produce a $\cos 3\theta$ term?

De Moivre $(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$.

Now expand the left side. Write $c = \cos\theta$, $s = \sin\theta$.

Expand

$(c + is)^3 = c^3 + 3ic^2s - 3cs^2 - is^3$.

Compare the two expressions.

Real parts

Equating real parts:

$\cos 3\theta = c^3 - 3cs^2$.

Use $s^2 = 1 - c^2$:

$\cos 3\theta = c^3 - 3c(1 - c^2)$

$= \boxed{4\cos^3\theta - 3\cos\theta}$.

Applications of de Moivre #1: Trig identities

Solution 29

Use De Moivre to express (a) $\cos 6\theta$, (b) $\sin 6\theta / \sin\theta$, in powers of $\cos\theta$. For (b), $\theta \neq k\pi$.

Set $c = \cos\theta$, $s = \sin\theta$.

$$(c+is)^6 = c^6 + 6ic^5 s - 15c^4 s^2 - 20ic^3 s^3 + 15c^2 s^4 + 6ics^5 - s^6.$$

a

Real parts, with $s^2 = 1 - c^2$:

$\cos 6\theta = c^6 - 15c^4 s^2 + 15c^2 s^4 - s^6$

$= 32c^6 - 48c^4 + 18c^2 - 1.$

b

Imaginary parts divided by s:

$$\frac{\sin 6\theta}{\sin\theta} = 6c^5 - 20c^3 s^2 + 6cs^4$$

$$= 32c^5 - 32c^3 + 6c.$$

Test Your Understanding

Solution 30

Edexcel FP2 June 2011 · Q7

Use De Moivre's theorem to show

$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta.$

(5 marks)

Given $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, solve $\sin 5\theta = 5\sin 3\theta$ for $0 \leq \theta < 2\pi$, to 3 decimal places. (6 marks)

Identity

Let $s = \sin\theta, c = \cos\theta$. Imaginary parts of $(c+is)^5$ give

$\sin 5\theta = 5c^4 s - 10c^2 s^3 + s^5$

$= 5(1-s^2)^2 s - 10(1-s^2)s^3 + s^5$

$= 16s^5 - 20s^3 + 5s.$

Solutions

$16s^5 - 20s^3 + 5s = 15s - 20s^3$

$2s(8s^4 - 5) = 0.$

Thus $s = 0$ or $s = \pm(5/8)^{1/4}$.

$\theta = 0,\ 1.095,\ 2.046,\ 3.142,\ 4.237,\ 5.188.$

Identities for powers of sine and cosine

Solution 31

Can we reverse the process: express powers of sine and cosine using multiple angles?

Let $z = \cos\theta + i\sin\theta$. Find $z + z^{-1}$ and $z - z^{-1}$.

First powers

$z^{-1} = \cos\theta - i\sin\theta.$

$z + z^{-1} = 2\cos\theta, \quad z - z^{-1} = 2i\sin\theta.$

Use De Moivre to generalise.

General powers

$z^n + z^{-n} = 2\cos n\theta,$

$z^n - z^{-n} = 2i\sin n\theta.$

These results can be derived when needed. Expand powers of $z + z^{-1}$ or $z - z^{-1}$, then group matching exponents.

Identities for powers of sine and cosine

Solution 32

Express $\cos^5\theta$ in terms of cosines of multiples of θ.

Let $z = e^{i\theta}$, so $z + z^{-1} = 2\cos\theta$.

Start $(z + z^{-1})^5 = 32\cos^5\theta.$

Expand

$z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5}.$

Group

$(z^5 + z^{-5}) + 5(z^3 + z^{-3}) + 10(z + z^{-1}).$

Substitute

$32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta.$

$$\boxed{\cos^5\theta = \frac{\cos 5\theta + 5\cos 3\theta + 10\cos\theta}{16}}.$$

The exponents descend in steps of 2. Pair equal positive and negative powers.

Identities for powers of sine and cosine

Solution 33

Express $\sin^3\theta$ in terms of sines of multiples of θ.

Use $z - z^{-1} = 2i\sin\theta$.

Start $(z - z^{-1})^3 = -8i\sin^3\theta.$

Expand

$z^3 - 3z + 3z^{-1} - z^{-3}.$

Group

$(z^3 - z^{-3}) - 3(z - z^{-1})$

$= 2i\sin 3\theta - 6i\sin\theta.$

Simplify

$$\boxed{\sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}}.$$

The binomial signs alternate. Dividing by $-8i$ changes the signs.

Test Your Understanding

Solution 34

(a) Express $\sin^4\theta$ in terms of cosines of multiples of θ.

(b) Hence evaluate $\int_0^{\pi/2} \sin^4\theta\, d\theta$.

Expand

$(z - z^{-1})^4 = 16\sin^4\theta.$

$= z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4}.$

a

$16\sin^4\theta = 2\cos 4\theta - 8\cos 2\theta + 6.$

$$\sin^4\theta = \frac{\cos 4\theta - 4\cos 2\theta + 3}{8}.$$

b

$$\int_0^{\pi/2} \sin^4\theta\, d\theta$$

$$= \frac{1}{8}\left[\frac{\sin 4\theta}{4} - 2\sin 2\theta + 3\theta\right]_0^{\pi/2} = \boxed{\frac{3\pi}{16}}.$$

Exercise 1D - Core Pure Year 2 Pages 14-15 — Solution 35

Exercise 1D · Question 2

(a) Use De Moivre's theorem to show

$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.

(b) Given $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, solve $\cos 5\theta + 5\cos 3\theta = 0$ for $0 \le \theta < \pi$, to 3 decimal places.

a

Put $c = \cos\theta$, $s = \sin\theta$. Compare real parts of $(c + is)^5$:

$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$.

Substitute $s^2 = 1 - c^2$, then expand to obtain $16c^5 - 20c^3 + 5c$.

b

$16c^5 - 10c = 2c(8c^4 - 5) = 0$.

Thus $c = 0$ or $c = \pm(5/8)^{1/4}$.

$\boxed{\theta = 0.475,\ 1.571,\ 2.666}$.

Exercise 1D - Core Pure Year 2 Pages 14-15 — Solution 36

Exercise 1D · Question 3

(a) Prove

$32\cos^6\theta = \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10$.

(6 marks)

(b) Evaluate $\int_0^{\pi/6} \cos^6\theta\, d\theta$. (3 marks)

a

Expand $(z + z^{-1})^6 = 64\cos^6\theta$:

$z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}$.

Pair opposite powers using $z^n + z^{-n} = 2\cos n\theta$, then divide by 2.

b

$$\frac{1}{32}\left[\frac{\sin 6\theta}{6} + \frac{3}{2}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/6}$$

$$= \boxed{\frac{5\pi}{96} + \frac{9\sqrt{3}}{64}}.$$

Exercise 1D · Question 4

(a) Prove

$32\cos^2\theta\sin^4\theta = \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2.$

(b) Evaluate $\int_0^{\pi/3} \cos^2\theta\sin^4\theta\,d\theta$.

a

$(z+z^{-1})^2(z-z^{-1})^4 = 64\cos^2\theta\sin^4\theta.$

Expanding gives

$z^6 - 2z^4 - z^2 + 4 - z^{-2} - 2z^{-4} + z^{-6}.$

Pair opposite powers and divide by 2.

b

$$\frac{1}{32}\left[\frac{\sin 6\theta}{6} - \frac{\sin 4\theta}{2} - \frac{\sin 2\theta}{2} + 2\theta\right]_0^{\pi/3}$$

The sine terms cancel, leaving $\boxed{\pi/48}$.

Exercise 1D · Question 6

(a) Show

$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1.$

(b) Find the six roots of

$32x^6 - 48x^4 + 18x^2 - \frac{3}{2} = 0$

to 3 decimal places.

a

Compare real parts of $(c+is)^6$:

$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6.$

Substitute $s^2 = 1 - c^2$ and simplify.

b

Put $x = \cos\theta$. Then $\cos 6\theta = 1/2$, so

$\theta = \pm\frac{\pi}{18} + \frac{k\pi}{3}.$

The six distinct values are

$\boxed{x = \pm 0.985,\ \pm 0.643,\ \pm 0.342}$.

There are six distinct roots of a degree-six polynomial, so these are all the roots.

Exercise 1D - Core Pure Year 2 Pages 14-15 — Solution 39

Exercise 1D · Question 7

(a) Show $\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$. (4 marks)

(b) Show

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}.$$

(4 marks)

(c) Solve $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$, to 2 decimal places. (5 marks)

a

Compare imaginary parts of $(c + is)^4$:
$\sin 4\theta = 4c^3 s - 4cs^3$.

b

The real part gives $\cos 4\theta = c^4 - 6c^2 s^2 + s^4$.
Divide the imaginary part by the real part, then divide numerator and denominator by c^4.

c

The quartic is equivalent to $\tan 4\theta = 1$, with $x = \tan\theta$.

$$\theta = \frac{\pi}{16} + \frac{k\pi}{4}, \quad k = 0, 1, 2, 3.$$

$$\boxed{x = 0.20,\ 1.50,\ -5.03,\ -0.67}.$$

Sums of Series — Solution 40

Complex geometric series

A geometric series starts $w, wz, wz^2, \ldots$. The common ratio is z.

Finite sum

$$S_n = \frac{w(1 - z^n)}{1 - z} \quad (z \neq 1).$$

If $z = 1$, $S_n = nw$.

Convergence

The modulus measures distance from zero: $|-4| = 4$, $|3 + 4i| = 5$.

Infinite sum

If $|z| < 1$, then $z^n \to 0$, and

$$S_\infty = \frac{w}{1 - z}.$$

A non-zero geometric series converges precisely when the modulus of its ratio is less than 1.

Example

Solution 41

For a positive integer n, let $z = e^{i\pi/n}$. Prove $1 + z + \cdots + z^{n-1} = 1 + i\cot\dfrac{\pi}{2n}$.

Answer

Let $\alpha = \pi/(2n)$. Since $z^n = -1$,

$$S = \frac{-2}{e^{2i\alpha} - 1} = \frac{-2}{2ie^{i\alpha}\sin\alpha}.$$

$$S = \frac{ie^{-i\alpha}}{\sin\alpha} = \frac{\sin\alpha + i\cos\alpha}{\sin\alpha}.$$

$$\boxed{S = 1 + i\cot\alpha}.$$

Let's practise that hard bit...

Solution 42

Half-angle rearrangements

Factor out the exponential with half the angle:

$e^{2i\alpha} - 1 = 2ie^{i\alpha}\sin\alpha$.

Simplify (1) $\dfrac{3}{e^{2i\theta} - 1}$.

1

$$\frac{3e^{-i\theta}}{2i\sin\theta}.$$

(2) $\dfrac{4e^{i\theta}}{e^{4i\theta} - 1}$

2

$$\frac{4e^{-i\theta}}{2i\sin 2\theta}.$$

(3) $\dfrac{1}{e^{i\theta/3} - 1}$

3

$$\frac{e^{-i\theta/6}}{2i\sin(\theta/6)}.$$

These formulas apply where the original denominators are non-zero.

Using mod-arg form to split summation

Solution 43

Using modulus–argument form to split a sum

For $\theta \neq 2k\pi$,

$S = e^{i\theta} + e^{2i\theta} + \cdots + e^{ni\theta}$

$= \dfrac{e^{i\theta}(e^{ni\theta} - 1)}{e^{i\theta} - 1}.$

The first term is $e^{i\theta}$, common ratio $e^{i\theta}$, and there are n terms.

Expand each exponential using Euler's formula:

$S = (\cos\theta + i\sin\theta) + \cdots$

$+ (\cos n\theta + i\sin n\theta).$

Group the real and imaginary parts:

$\operatorname{Re} S = \cos\theta + \cdots + \cos n\theta,$

$\operatorname{Im} S = \sin\theta + \cdots + \sin n\theta.$

Thus one complex geometric sum gives both trigonometric sums.

Example

Solution 44

Let

$P = \cos\theta + \cos 2\theta + \cdots + \cos 8\theta,$

$Q = \sin\theta + \sin 2\theta + \cdots + \sin 8\theta.$

(a) Show

$P + iQ = e^{9i\theta/2}\dfrac{\sin 4\theta}{\sin(\theta/2)}.$

Assume $\sin(\theta/2) \neq 0$.

Series

$P + iQ = e^{i\theta} + e^{2i\theta} + \cdots + e^{8i\theta}.$

This is geometric with first term and ratio $e^{i\theta}$.

$P + iQ = e^{i\theta}\dfrac{e^{8i\theta} - 1}{e^{i\theta} - 1}.$

Simplify

$= e^{i\theta}\dfrac{2ie^{4i\theta}\sin 4\theta}{2ie^{i\theta/2}\sin(\theta/2)}$

$= e^{9i\theta/2}\dfrac{\sin 4\theta}{\sin(\theta/2)}.$

Example

Solution 45

For the sums P, Q on the previous slide,

$$P + iQ = e^{9i\theta/2}\frac{\sin 4\theta}{\sin(\theta/2)}.$$

(b) Find expressions for P and Q, and simplify Q/P.

P

Compare real parts:

$$P = \cos\frac{9\theta}{2}\sin 4\theta \csc\frac{\theta}{2}.$$

Q

Compare imaginary parts:

$$Q = \sin\frac{9\theta}{2}\sin 4\theta \csc\frac{\theta}{2}.$$

Ratio

Where $P \neq 0$,

$$\boxed{Q/P = \tan(9\theta/2)}.$$

Exercise 1E - Core Pure Year 2 Pages 18-19

Solution 46

Exercise 1E · Question 4

$$C = 1 + \frac{\cos\theta}{3} + \frac{\cos 2\theta}{9} + \cdots,$$

$$S = \frac{\sin\theta}{3} + \frac{\sin 2\theta}{9} + \cdots.$$

(a) Show $C + iS = 3/(3 - e^{i\theta})$.

(b) Find real expressions for C, S.

a

$C + iS$ is geometric with ratio $e^{i\theta}/3$, whose modulus is $1/3 < 1$.

$$C + iS = \frac{1}{1 - e^{i\theta}/3} = \frac{3}{3 - e^{i\theta}}.$$

b

Multiply by $3 - e^{-i\theta}$:

$$C + iS = \frac{9 - 3\cos\theta + 3i\sin\theta}{10 - 6\cos\theta}.$$

$$\boxed{C = \frac{9 - 3\cos\theta}{10 - 6\cos\theta}, \quad S = \frac{3\sin\theta}{10 - 6\cos\theta}}.$$

Exercise 1E · Question 5

For $0 < \theta < \pi$,

$P = 1 + \cos\theta + \cdots + \cos 12\theta$,

$Q = \sin\theta + \cdots + \sin 12\theta$.

(a) Show

$$P + iQ = e^{6i\theta}\frac{e^{13i\theta/2} - e^{-13i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}}.$$

(4 marks)

(b) Deduce real expressions for P, Q. (4 marks)

a

$P + iQ = \dfrac{e^{13i\theta} - 1}{e^{i\theta} - 1}$.

Factor $e^{13i\theta/2}$ from the numerator and $e^{i\theta/2}$ from the denominator. Their quotient is $e^{6i\theta}$, giving the required expression.

b

Use $e^{ix} - e^{-ix} = 2i\sin x$:

$$P + iQ = e^{6i\theta}\sin\frac{13\theta}{2}\csc\frac{\theta}{2}.$$

$$P = \cos 6\theta \sin\frac{13\theta}{2}\csc\frac{\theta}{2},$$

$$Q = \sin 6\theta \sin\frac{13\theta}{2}\csc\frac{\theta}{2}.$$

Exercise 1E · Question 6

$$C = \sum_{k=0}^{n}\binom{n}{k}\cos k\theta, \qquad S = \sum_{k=1}^{n}\binom{n}{k}\sin k\theta.$$

(a) Show $C = (2\cos(\theta/2))^n \cos(n\theta/2)$.

(b) Show $S/C = \tan(n\theta/2)$, where defined.

a

By the binomial theorem,

$C + iS = (1 + e^{i\theta})^n$.

$1 + e^{i\theta} = 2e^{i\theta/2}\cos(\theta/2)$.

Thus

$C + iS = (2\cos(\theta/2))^n e^{in\theta/2}$.

Take real parts to obtain (a).

b

The imaginary part is

$S = (2\cos(\theta/2))^n \sin(n\theta/2)$.

Divide by $C \neq 0$: $\boxed{S/C = \tan(n\theta/2)}$.

Exercise 1E - Core Pure Year 2 Pages 18-19 — Solution 49

Exercise 1E · Question 7

(a) Show $(2+e^{i\theta})(2+e^{-i\theta}) = 5+4\cos\theta$. (2 marks)

$$C = 1 - \frac{\cos\theta}{2} + \frac{\cos 2\theta}{4} - \cdots,$$

$$S = \frac{\sin\theta}{2} - \frac{\sin 2\theta}{4} + \cdots.$$

(b) By considering $C - iS$, find C, S. (4 marks)

a

Expand: $4 + 2(e^{i\theta} + e^{-i\theta}) + 1 = 5 + 4\cos\theta$.

b

$$C - iS = \frac{1}{1 + e^{i\theta}/2} = \frac{2}{2 + e^{i\theta}}.$$

Multiply by $2 + e^{-i\theta}$:

$$C - iS = \frac{4 + 2\cos\theta - 2i\sin\theta}{5 + 4\cos\theta}.$$

$$\boxed{C = \frac{4 + 2\cos\theta}{5 + 4\cos\theta}, \quad S = \frac{2\sin\theta}{5 + 4\cos\theta}.}$$

Applications of de Moivre #2: Roots — Solution 50

Cube roots of unity

Solve $z^3 = 1$. Write $z = re^{i\theta}$.

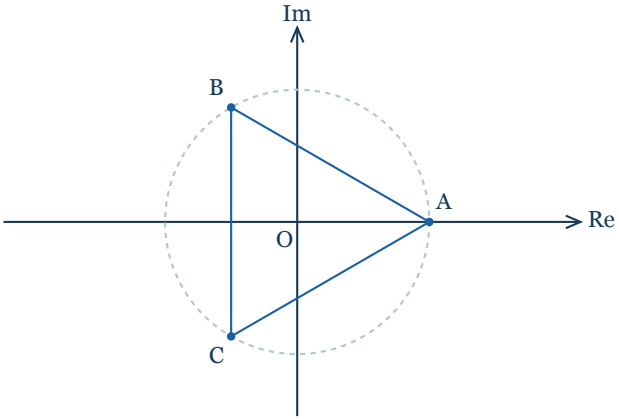
Modulus and argument

$r^3 = 1$, so $r = 1$.

$3\theta = 2k\pi, \quad \theta = \frac{2k\pi}{3}$.

Roots

$$\boxed{z = 1,\ -\frac{1}{2} + \frac{\sqrt{3}}{2}i,\ -\frac{1}{2} - \frac{\sqrt{3}}{2}i.}$$



The roots lie on the unit circle. Consecutive arguments differ by $2\pi/3$, so they form an equilateral triangle.

More on Roots of Unity

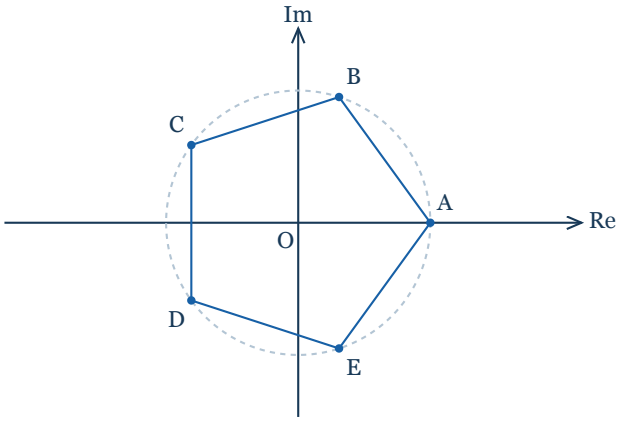
Solution 51

The nth roots of unity

$z^n = 1 \implies z = e^{2k\pi i/n}, \quad k = 0, \ldots, n-1.$

They are equally spaced on the unit circle, with angular separation $2\pi/n$.

Put $\omega = e^{2\pi i/n}$. The roots are $1, \omega, \ldots, \omega^{n-1}$.



For $n \geq 2$, show that their sum is zero.

Sum

$1 + \omega + \cdots + \omega^{n-1} = \dfrac{1-\omega^n}{1-\omega} = 0.$

Here $\omega^n = 1$ and $\omega \neq 1$.

Geometrically, the vectors are balanced around the origin.

Example

Solution 52

Find all four roots of

$z^4 = 2 + 2\sqrt{3}i$

in exponential form, using principal arguments.

Answer

$2 + 2\sqrt{3}i = 4e^{i\pi/3}.$

Thus $|z| = \sqrt{2}$, and

$4\theta = \dfrac{\pi}{3} + 2k\pi \implies \theta = \dfrac{\pi}{12} + \dfrac{k\pi}{2}.$

Choose four distinct principal arguments:

$$\boxed{z = \sqrt{2}e^{i\theta}, \quad \theta = \frac{\pi}{12}, \frac{7\pi}{12}, -\frac{11\pi}{12}, -\frac{5\pi}{12}}.$$

Test Your Understanding — Solution 53

Find the modulus and argument of $-2+2\sqrt{3}i$.

Hence solve $z^4 = -2+2\sqrt{3}i$, using exponential form and principal arguments.

a

$|-2+2\sqrt{3}i| = 4$, $\arg(-2+2\sqrt{3}i) = 2\pi/3$.

b

$|z| = \sqrt{2}$, and $\theta = \pi/6 + k\pi/2$.

$$\boxed{z = \sqrt{2}e^{i\theta}, \quad \theta = \frac{\pi}{6}, \frac{2\pi}{3}, -\frac{5\pi}{6}, -\frac{\pi}{3}}.$$

Exercise 1F - Core Pure Year 2 Pages 24-25 — Solution 54

Exercise 1F · Question 4

(a) Find the three roots of $(z+1)^3 = -1$, in the form $x+iy$.

(b) Plot the roots on an Argand diagram.

(c) Find the centre and radius of the circle through them.

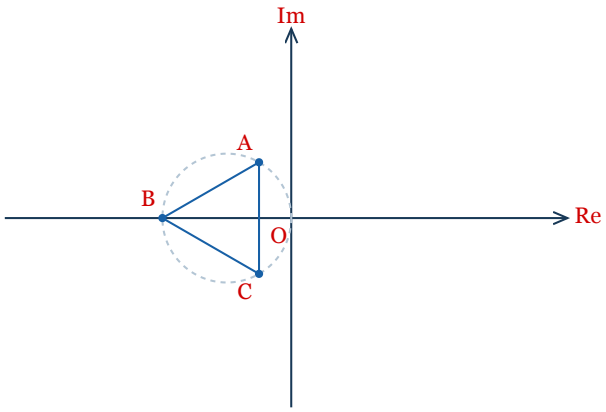
a

$z+1 = e^{i(\pi+2k\pi)/3}$.

For $k = 0, 1, 2$:

$$\boxed{z = -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i,\ -2,\ -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i}.$$

Argand diagram



c

The roots of $w^3 = -1$ lie on the unit circle. Since $z = w - 1$, translate the circle one unit left.

Centre $\boxed{(-1, 0)}$; radius $\boxed{1}$.

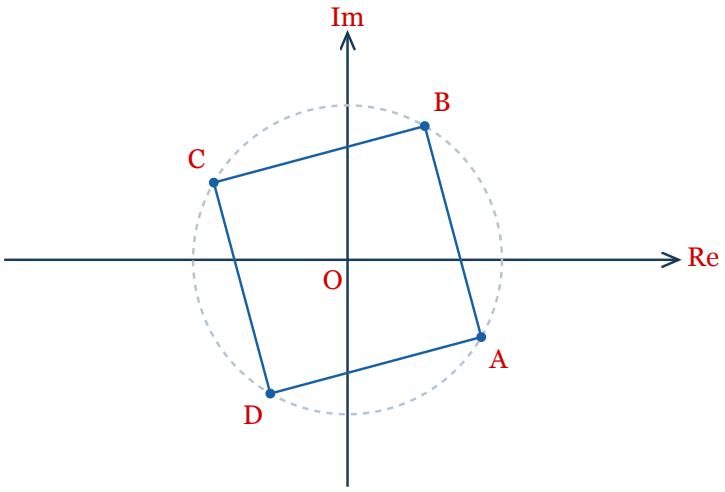
Exercise 1F · Question 6

(a) Find the modulus and argument of $-2-2i\sqrt{3}$.

(b) Solve $z^4+2+2i\sqrt{3}=0$. Give $re^{i\theta}$, $r>0$, $-\pi<\theta\le\pi$, and illustrate the roots.

a $r=4,\ \theta=-2\pi/3.$

Argand diagram



b

$z^4=4e^{-2i\pi/3}, \quad |z|=\sqrt{2}.$

$\arg z=-\dfrac{\pi}{6}+\dfrac{k\pi}{2}.$

The four principal arguments are

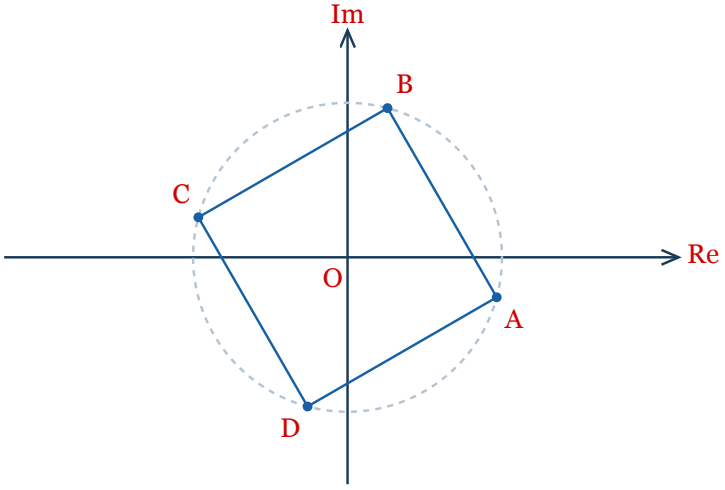
$$\boxed{-\frac{2\pi}{3},-\frac{\pi}{6},\frac{\pi}{3},\frac{5\pi}{6}}.$$

Each root is $\sqrt{2}e^{i\theta}$; adjacent roots are $\pi/2$ apart.

Exercise 1F · Question 7

Find the four distinct roots of $z^4=2(1-i\sqrt{3})$ in exponential form, and show them on an Argand diagram. (7 marks)

Argand diagram



Answer

$2(1-i\sqrt{3})=4e^{-i\pi/3}.$

Therefore $|z|=\sqrt{2}$ and

$\theta=-\dfrac{\pi}{12}+\dfrac{k\pi}{2}.$

The roots are $\sqrt{2}e^{i\theta}$, where

$$\boxed{\theta=-\frac{7\pi}{12},-\frac{\pi}{12},\frac{5\pi}{12},\frac{11\pi}{12}}.$$

Exercise 1F · Question 8

$z = \sqrt{6} + i\sqrt{2}$.

(a) Find $|z|$ and $\arg z$. (2 marks)

(b) Find all w such that $w^3 = z^4$, in exponential form with principal arguments. (4 marks)

a

$|z| = \sqrt{8} = 2\sqrt{2}, \quad \arg z = \pi/6.$

b

$w^3 = (2\sqrt{2})^4 e^{2i\pi/3} = 64e^{2i\pi/3}.$

Hence $|w| = 4$, and $\arg w = 2\pi/9 + 2k\pi/3$.

$$\boxed{w = 4e^{2i\pi/9},\ 4e^{8i\pi/9},\ 4e^{-4i\pi/9}}.$$

Exercise 1F · Question 9

(a) Solve

$1 + z + z^2 + \cdots + z^7 = 0.$

(b) Hence deduce that $z^2 + 1$ and $z^4 + 1$ are factors of this polynomial.

a

At $z = 1$ the sum is 8, so $z \neq 1$. The geometric sum gives

$$\frac{z^8 - 1}{z - 1} = 0.$$

Thus the solutions are the eighth roots of unity excluding 1:

$$\boxed{z = e^{ik\pi/4}, \quad k = 1, \ldots, 7}.$$

b

$z^8 - 1 = (z - 1)(z + 1)(z^2 + 1)(z^4 + 1).$

Dividing by $z - 1$ proves both required factors. Equivalently, each root of either factor is an eighth root of unity other than 1.

Challenge

(a) Find the six roots of $z^6 = 1$ as $e^{i\theta}$, with principal arguments.

(b) Hence show that the solutions of $(z+1)^6 = z^6$ are $z = -\frac{1}{2} + \frac{i}{2}\cot\frac{k\pi}{6}, \quad k = 1, \ldots, 5.$

a

$\theta = -2\pi/3, -\pi/3, 0, \pi/3, 2\pi/3, \pi$.

b

Neither $z = 0$ nor $z = -1$ is a solution. Put $u = z/(z+1)$. Then $u^6 = 1$, but $u \neq 1$.

Write $u = e^{2i\alpha}$, $\alpha = k\pi/6$, $k = 1, \ldots, 5$.

$$z = \frac{u}{1-u} = \frac{e^{i\alpha}}{-2i\sin\alpha}$$

$$= \frac{i(\cos\alpha + i\sin\alpha)}{2\sin\alpha} = \boxed{-\frac{1}{2} + \frac{i}{2}\cot\alpha}.$$

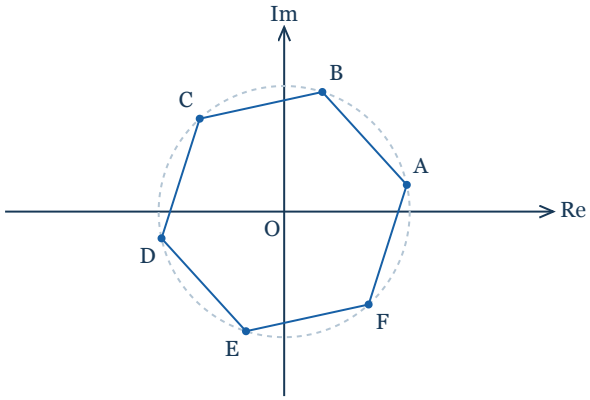
Roots and regular polygons

If z_1 is one root of $z^n = w \neq 0$, all roots are

$z_1, z_1\omega, \ldots, z_1\omega^{n-1}, \quad \omega = e^{2\pi i/n}$.

They form a regular n-gon centred at the origin.

For example, the roots of $z^6 = 7 + 24i$ have modulus $25^{1/6}$ and angular separation $\pi/3$.



Multiplication by ω rotates a point by $2\pi/n$ without changing its distance from the origin.

Once one vertex is known, rotate repeatedly to generate the others.

Example

Solution 61

An equilateral triangle is centred at the origin. One vertex is $P(\sqrt{3}, 1)$.

Find the other two vertices and the area.

Vertices

P represents $2e^{i\pi/6}$. Rotate by $2\pi/3$ and $4\pi/3$.

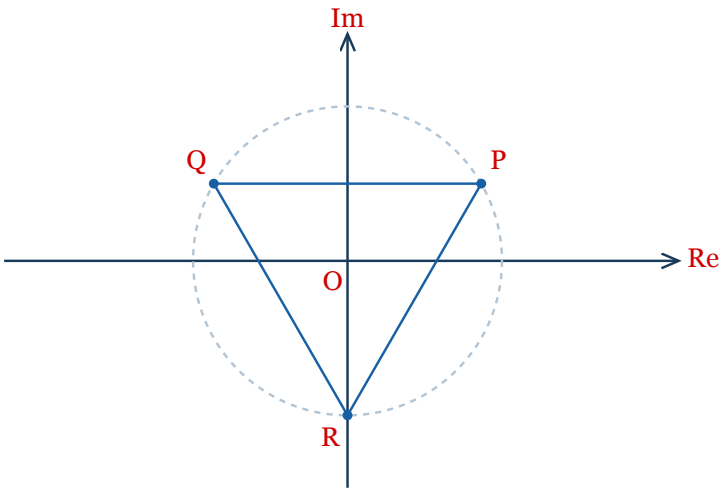
$$\boxed{Q(-\sqrt{3}, 1), \quad R(0, -2)}.$$

Area

Base $PQ = 2\sqrt{3}$, height $= 3$.

$$A = \frac{1}{2}(2\sqrt{3})(3) = \boxed{3\sqrt{3}}.$$

Argand diagram



Exercise 1G - Core Pure Year 2 Pages 26-27

Solution 62

Exercise 1G · Question 3

Triangle OAB is equilateral on an Argand diagram. O is the origin; A represents $\sqrt{3}(1 - i)$, and B represents b.

Find both possibilities for b in exponential form and sketch both triangles. (5 marks)

Answer

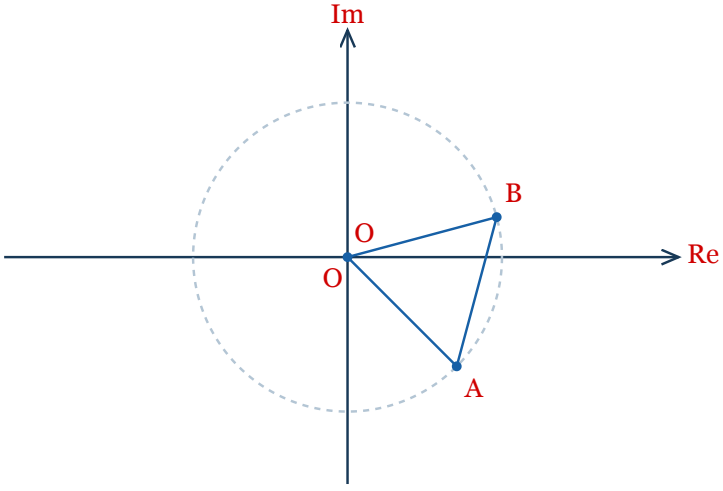
$a = \sqrt{6}e^{-i\pi/4}$.

Rotate OA by $\pm\pi/3$:

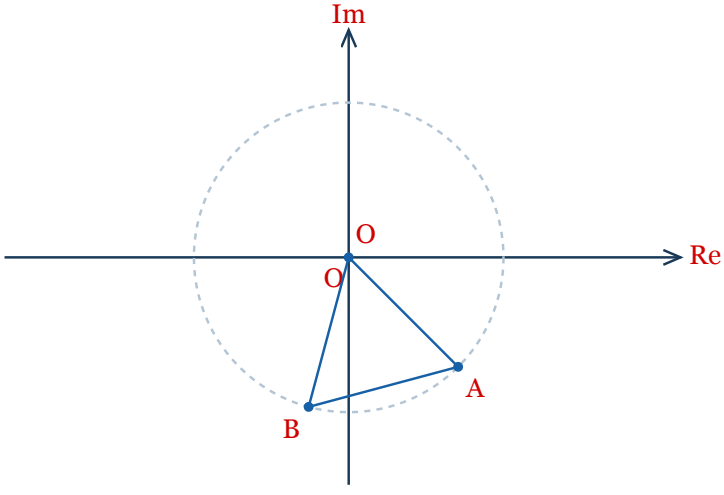
$b = ae^{\pm i\pi/3}$.

$$\boxed{b = \sqrt{6}e^{i\pi/12} \text{ or } \sqrt{6}e^{-7i\pi/12}}.$$

Argand diagram



Second triangle



Exercise 1G · Question 4

(a) Find the fourth roots of $-12i$, as $re^{i\theta}$ with principal arguments. Illustrate them. (6 marks)

In increasing argument order, label them A, B, C, D.

(b) Their side midpoints are the fourth roots of a complex number w. Find w. (4 marks)

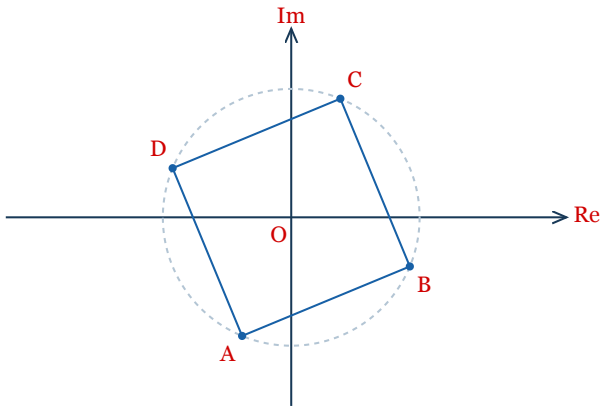
a

$$z = \sqrt[4]{12}e^{i\theta}, \quad \theta = -\frac{\pi}{8} + \frac{k\pi}{2}.$$

In increasing order:

$$\boxed{\theta = -5\pi/8, -\pi/8, 3\pi/8, 7\pi/8}.$$

Argand diagram



b

The roots form a square. A side midpoint has modulus $\sqrt[4]{12}/\sqrt{2} = \sqrt[4]{3}$.

The midpoint of AB has argument $-3\pi/8$. Therefore

$$w = \left(\sqrt[4]{3}e^{-3i\pi/8}\right)^4 = 3e^{-3i\pi/2} = \boxed{3i}.$$

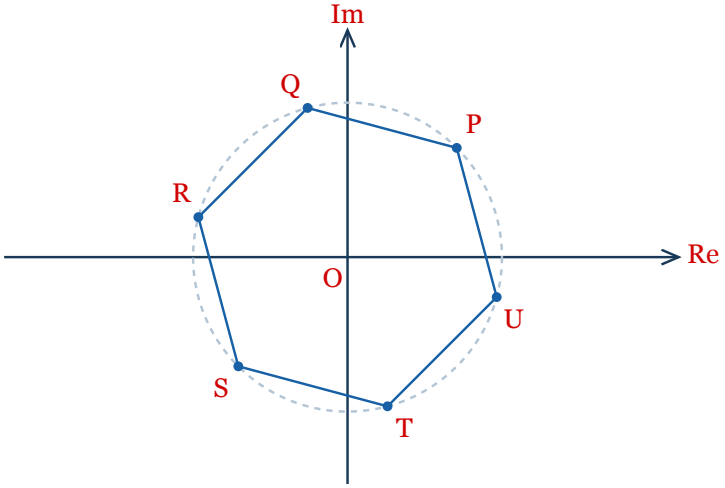
Exercise 1G · Question 5

A regular hexagon on an Argand diagram is centred at the origin. Vertex P represents $8 + 8i$.

Find the other five vertices in the form $a + bi$, and illustrate them.

Use $z_k = (8 + 8i)e^{ik\pi/3}$.

Argand diagram



Answer

For $k = 1, \ldots, 5$, the vertices are

$4(1 - \sqrt{3}) + 4(1 + \sqrt{3})i$,

$-4(1 + \sqrt{3}) + 4(\sqrt{3} - 1)i$,

$-8 - 8i$,

$4(\sqrt{3} - 1) - 4(1 + \sqrt{3})i$,

$4(1 + \sqrt{3}) + 4(1 - \sqrt{3})i$.

They lie on the circle of radius $8\sqrt{2}$, separated by $\pi/3$.